

The Probabilistic Foundations of Asymmetric Interval Numbers

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Abstract. Classical intervals represent uncertainty through bounds alone and, under the principle of indifference, correspond to the uniform distribution on an interval. In many applications, however, additional information is available in the form of an expected value within the bounds, producing an asymmetric pattern of uncertainty that classical intervals cannot represent. We provide a unified probabilistic foundation for Asymmetric Interval Numbers (AINs). An AIN, defined by the triple (a, b, c) of lower bound, upper bound, and expected value, induces a piecewise-uniform probability density uniquely determined by the normalization and expectation constraints; the classical interval appears as the symmetric special case. We prove the uniqueness of this induced distribution and derive closed-form expressions for its variance and differential entropy, completing the distributional characterization of AINs. The resulting construction interprets AINs as an information-consistent representation of partial knowledge rather than as an assumed stochastic model. Consequently, AINs extend classical interval arithmetic by incorporating expected-value information while preserving analytical tractability. A reference implementation is available in the open-source **asymintervals** library.

Keywords: Interval numbers · Asymmetric Interval Number · AIN.

1 Introduction

In many measurement and decision problems, quantities are not known precisely but are specified through bounds. The classical interval $[a, b]$, introduced by Moore [3], formalizes the statement that an unknown value x satisfies $a \leq x \leq b$. When no further information is available, the principle of indifference implies that no value within the interval should be favored over any other, yielding the uniform distribution $\mathcal{U}(a, b)$ with midpoint $(a + b)/2$ as the natural point estimate.

However, practitioners often possess more information than bounds alone. A driver may report fuel consumption between 7 and 19 liters per 100km, with a

typical value of 10.5. The interval $[7, 19]$ captures the range but its midpoint, 13, substantially overestimates the expected consumption. The exact distribution is unknown, yet the available information, namely the minimum, the maximum, and the expected value, clearly indicates an asymmetric pattern that the classical interval cannot represent.

Asymmetric Interval Numbers (AINs), introduced in [5], address this by augmenting the interval with the expected value $c \in [a, b]$ satisfying $\mathbb{E}(X) = c$. An AIN $X = [a, b]_c$ represents an uncertain quantity in $[a, b]$ with $\mathbb{E}(X) = c$. The value c divides $[a, b]$ into two subintervals. Within each subinterval, no information distinguishes one value from another, so by the principle of indifference the density is constant on each piece. Since the two constraints, normalization and $\mathbb{E}(X) = c$, uniquely determine the two density levels, no free parameters remain: the distribution is not chosen but determined by the available information. This construction may also be interpreted as selecting the least informative piecewise-constant distribution consistent with the constraints given. The resulting piecewise-uniform distribution is therefore not an arbitrary modeling choice; it is the logical consequence of applying the same reasoning that justifies $\mathcal{U}(a, b)$ for classical intervals, extended by one additional piece of information.

Since their introduction, AINs have been developed in several directions, including algebraic properties [5], software implementations [4, 6] and applications [8, 9]. However, these developments are spread across a series of contributions and are not accompanied by a single, unified account of what AINs are and what they are not. In particular, the foundational justification of the piecewise-uniform density and the link between the AIN construction and the classical uniform model associated with intervals have not been explicitly stated in one place [7].

The primary aim of this paper is to provide such a unified foundation. We consolidate the existing definitions and arithmetic within a coherent probabilistic framework, clarifying that the AIN distribution is not an arbitrary modeling assumption, but a logical consequence of applying the principle of indifference to the two subintervals induced by c . The resulting construction yields an information-consistent representation of partial knowledge rather than a model of any physical data-generating process.

The contributions of this paper are as follows. We consolidate the definitions, arithmetic, and distributional properties of AINs, previously spread across several publications, within a single coherent probabilistic framework. In the course of this consolidation, we provide a formal uniqueness proof for the piecewise-uniform density (Theorem 1), a property previously only implicit in the AIN literature, and derive new closed-form expressions for the variance (Proposition 1) and differential entropy (Eq. 19). We further show that classical interval numbers arise as a symmetric special case (Theorem 2), so the framework extends classical interval arithmetic rather than replacing it.

The remainder of the paper is organized as follows. Section 2 recalls the classical interval framework and derives the AIN distribution from the principle of indifference, establishing its uniqueness and its reduction to the classical interval

in the symmetric case. Section 3 develops arithmetic operations on AINs using the Law of the Unconscious Statistician. Section 4 presents the reference software implementation. Section 5 analyzes the distributional properties of AINs, including the cumulative distribution function, variance, and entropy. Section 6 discusses open problems and possible extensions of the framework. Section 7 concludes the paper.

2 From Intervals to Asymmetric Interval Numbers

This section recalls the classical interval framework introduced by Moore and shows how the principle of indifference, combined with one additional piece of information, naturally leads to the AIN distribution.

2.1 Classical Interval Numbers

A classical interval number (CIN) is a closed bounded subset of the real line:

$$X = [a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}, \quad a \leq b. \quad (1)$$

The interval formalizes the statement that the value of an uncertain quantity x is known only to satisfy $a \leq x \leq b$. No further information about x is assumed.

Under the principle of indifference [2], the absence of any reason to favor one value over another within $[a, b]$ implies a constant probability density over the interval. The resulting probability distribution is the uniform distribution $\mathcal{U}(a, b)$ with density

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

The expected value of this distribution is the midpoint:

$$\mathbb{E}(X) = \frac{a+b}{2} = \text{mid}(X). \quad (3)$$

The midpoint is also the minimax point estimate under the absolute loss function: it minimizes the worst-case absolute error $\max_{x \in [a, b]} |x - \hat{x}|$. Both the probabilistic argument (Laplace) and the minimax argument thus select the same value.

Binary operations on CINs are defined as set operations [3]. For $\circ \in \{+, -, \cdot, \div\}$ and two intervals $X = [a_1, b_1]$, $Y = [a_2, b_2]$:

$$X \circ Y = \{x \circ y \mid x \in X, y \in Y\}. \quad (4)$$

2.2 Introducing the Expected Value

In many practical situations, more is known about an uncertain quantity than its bounds alone. A practitioner may be able to specify the expected (typical, average) value c in addition to the minimum a and maximum b . The AIN representation captures this information.

Definition 1 (Asymmetric Interval Number). *An Asymmetric Interval Number (AIN) is the triple*

$$X = [a, b]_c, \quad (5)$$

where $a, b, c \in \mathbb{R}$ satisfy $a \leq c \leq b$. An AIN $X = [a, b]_c$ represents an uncertain quantity known to lie in $[a, b]$ with expected value $c = \mathbb{E}(X)$. An AIN with $a < b$ is called non-degenerate; an AIN with $a = b = c$ is called degenerate and corresponds to a known real number.

We associate each AIN with a probability measure on $[a, b]$, determined according to the principle of indifference and having expected value c . The nature of this measure depends on the triple (a, b, c) :

- For $a < c < b$, the piecewise-constant density construction of Section 2.3 yields an absolutely continuous distribution uniquely determined by the normalization and expectation constraints.
- For $c = a$ or $c = b$ (including the degenerate case $a = b = c$), the associated measure is the Dirac measure at c . These cases fall outside the piecewise-constant density framework and are not treated further in this paper.

2.3 Derivation from the Principle of Indifference

The derivation proceeds in three steps: identifying the partition induced by the available information, applying the principle of indifference within each part, and solving the resulting constraint equations.

The expected value c is the only distinguished point in the interior of $[a, b]$. It partitions the interval into two subintervals: $[a, c]$ (values below the mean) and $[c, b]$ (values at or above the mean). Since c is the only location at which the available information changes, the density may change only at c .

Within $[a, c]$, no information is available to distinguish one value from another. By the principle of indifference, the density must be constant on $[a, c]$. The same reasoning applies to $[c, b]$: the density must be constant on $[c, b]$. Denote these two density levels by α and β , respectively:

$$f_X(x) = \begin{cases} \alpha, & a \leq x < c, \\ \beta, & c \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The density f_X must satisfy two conditions. The first is the normalization constraint:

$$\int_a^b f_X(x) dx = \alpha(c - a) + \beta(b - c) = 1. \quad (7)$$

The second is the expectation constraint:

$$\int_a^b x f_X(x) dx = \alpha \frac{c^2 - a^2}{2} + \beta \frac{b^2 - c^2}{2} = c. \quad (8)$$

This yields a linear system of two equations in the unknowns (α, β) .

Theorem 1 (Uniqueness). *For any $a, b, c \in \mathbb{R}$ with $a < c < b$, the system (7)–(8) has a unique solution:*

$$\alpha = \frac{b - c}{(b - a)(c - a)}, \quad \beta = \frac{c - a}{(b - a)(b - c)}. \quad (9)$$

Moreover, $\alpha > 0$ and $\beta > 0$.

Proof. The system can be written in matrix form as

$$\begin{pmatrix} c - a & b - c \\ \frac{c^2 - a^2}{2} & \frac{b^2 - c^2}{2} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 \\ c \end{pmatrix}.$$

The determinant of the coefficient matrix is

$$W = (c - a) \frac{b^2 - c^2}{2} - (b - c) \frac{c^2 - a^2}{2} = \frac{(c - a)(b - c)(b - a)}{2}.$$

Since $a < c < b$, all three factors $(c - a)$, $(b - c)$, and $(b - a)$ are strictly positive, so $W > 0$. By Cramer's rule, the system has a unique solution. Solving yields (9). Since $b > c > a$, all factors in the numerators and denominators are positive, hence $\alpha > 0$ and $\beta > 0$. $\square$

The choice of the piecewise-constant class follows directly from the principle of indifference: the density may change only at c , the sole point where the available information changes. More expressive families, such as triangular or truncated exponential densities, would require parameters not supplied by the triple (a, b, c) . Within the chosen class, the density is not a modeling decision but a consequence of the constraints (Theorem 1). The relationship to entropy maximization is discussed in Section 5.2.

Remark 1. The uniqueness result means that within the class of piecewise-constant densities with breakpoint at c , the AIN distribution is fully determined by the constraints. There are no free parameters to choose, and therefore no modeling decision beyond the specification of (a, b, c) .

2.4 Symmetry, Asymmetry, and the Classical Case

When the expected value coincides with the midpoint of the interval, the AIN reduces to a classical interval number.

Theorem 2 (Reduction to $\mathcal{U}(a, b)$). *Let $X = [a, b]_c$ with $c = (a + b)/2$. Then $\alpha = \beta = 1/(b - a)$, and the AIN distribution reduces to the uniform distribution $\mathcal{U}(a, b)$.*

Proof. Substituting $c = (a + b)/2$ into (9) gives equal numerators $b - c = c - a = (b - a)/2$ and equal denominators, so $\alpha = \beta = 1/(b - a)$. $\square$

The classical interval numbers thus form a symmetric subclass of AINs. Any result valid for AINs in general automatically holds for CINs as a special case.

The departure from the symmetric case is measured by the asymmetry coefficient

$$A = \frac{a + b - 2c}{b - a}, \quad (10)$$

which takes values in the open interval $(-1, 1)$ for non-degenerate AINs. The value $A = 0$ corresponds to the classical interval (c at the midpoint). Positive asymmetry ($A > 0$) means that c is closer to the lower bound a , so that the density on $[a, c]$ is higher than on $[c, b]$. Negative asymmetry ($A < 0$) means that c is closer to the upper bound b , with higher density on $[c, b]$.

As $A \rightarrow 1^-$ (i.e., $c \rightarrow a^+$) or $A \rightarrow -1^+$ (i.e., $c \rightarrow b^-$), the distribution degenerates: one density level diverges while the corresponding subinterval shrinks to zero width. These boundary cases are excluded from the standard AIN definition. The case $a = b = c$ corresponds to a known real number; the asymmetry coefficient A is undefined for this degenerate case, as the formula (10) involves division by zero. In the software implementation, A is conventionally set to 0 for degenerate AINs, reflecting the absence of asymmetry.

3 Arithmetic Operations on AINs

Arithmetic on AINs extends classical interval arithmetic by propagating the expected value through operations. An arithmetic operation on AINs produces a new AIN whose bounds are determined by the range of the operation (as in Moore's arithmetic) and whose expected value is determined by the Law of the Unconscious Statistician (LOTUS) [1] applied to the induced piecewise-uniform density.

For a measurable function $g: \mathbb{R} \rightarrow \mathbb{R}$ and an AIN $X = [a, b]_c$, the expected value of the transformed quantity is

$$\mathbb{E}[g(X)] = \int_a^b g(x) f_X(x) dx = \alpha \int_a^c g(x) dx + \beta \int_c^b g(x) dx. \quad (11)$$

Because the density is piecewise constant, the integral in (11) reduces to two standard integrals. Whenever g has a closed-form antiderivative, the expected value of the result is also in closed form.

If g is monotonic on $[a, b]$, the transformed AIN takes the general form

$$g(X) = [\min(g(a), g(b)), \max(g(a), g(b))]_{\mathbb{E}[g(X)]}. \quad (12)$$

For monotonic g , the expectation necessarily lies strictly within the image interval, hence the result is again a valid AIN.

For binary operations on two AINs $X = [a_1, b_1]_{c_1}$ and $Y = [a_2, b_2]_{c_2}$, the expected value of the result is computed under the product of the two induced densities:

$$\mathbb{E}[g(X, Y)] = \int_{a_1}^{b_1} \int_{a_2}^{b_2} g(x, y) f_X(x) f_Y(y) dx dy. \quad (13)$$

Table 1. Arithmetic operations on AINs. Here $X = [a_1, b_1]_{c_1}$ and $Y = [a_2, b_2]_{c_2}$. Notation: $m_o = \min$, $M_o = \max$ over the four endpoint products or quotients. The density parameters α_i, β_i refer to the respective operand.

Operation	Bounds	Expected value
$X + Y$	$[a_1 + a_2, b_1 + b_2]$	$c_1 + c_2$
$X - Y$	$[a_1 - b_2, b_1 - a_2]$	$c_1 - c_2$
$X \cdot Y$	$[m_\cdot, M_\cdot]$	$c_1 \cdot c_2$
$X \div Y$	$[m_\div, M_\div]$	$c_1 \cdot (\alpha_2 \ln \frac{c_2}{a_2} + \beta_2 \ln \frac{b_2}{c_2})$
$\sqrt{X}$ ($a_1 \geq 0$)	$[\sqrt{a_1}, \sqrt{b_1}]$	$\frac{2}{3}(\alpha_1(c_1^{3/2} - a_1^{3/2}) + \beta_1(b_1^{3/2} - c_1^{3/2}))$
X^n	$[a_1^n, b_1^n]$	$\frac{1}{n+1}(\alpha_1(c_1^{n+1} - a_1^{n+1}) + \beta_1(b_1^{n+1} - c_1^{n+1}))$

The product form $f_X \cdot f_Y$ is adopted as the least-informative joint extension consistent with the marginal densities of X and Y . Since the AIN representation specifies only marginal information (a_i, c_i, b_i) for each operand, no joint structure is available, and the product density adds no information beyond what is contained in the individual operands. This is consistent with the principle of indifference applied to the joint domain. Note that for addition and subtraction, the expected value of the result equals $c_1 + c_2$ (resp. $c_1 - c_2$) by linearity of expectation, regardless of any joint assumption. For multiplication, $\mathbb{E}(XY) = c_1 c_2$ is a consequence of the product form and may not hold under dependence.

Table 1 summarizes the arithmetic operations defined for AINs. For addition and subtraction, the expected value propagates linearly and no integral evaluation is needed. For all other operations, the expected value is obtained via LOTUS. All listed operations admit analytic expressions. Complete derivations and proofs are given in [5].

The additive propagation of the expected value for addition and subtraction, and the multiplicative propagation for multiplication under the canonical joint extension (Table 1), make AIN arithmetic structurally consistent with the algebra of expectations.

4 Software Implementation

The theoretical framework described in the preceding sections is realized in the open-source Python library `asymintervals` [4,6], which serves as the reference implementation of AIN arithmetic. The library is available on PyPI and GitHub.³

The core object is the AIN class, constructed directly from the triple (a, b, c) . The argument order follows the interval notation: lower bound, upper bound, expected value. The constructor computes α and β from Eq. (9) and the asymmetry coefficient from Eq. (10). The methods `pdf()`, `cdf()`, and `quantile()`

³ <https://github.com/wsalabun/asymintervals>

evaluate the induced density, distribution function, and its inverse, respectively. Operator overloading (+, -, *, /, **) implements the arithmetic of Section 3, evaluating the analytic expressions of Table 1 exactly, without sampling or numerical integration. The reduction property (Theorem 2) is verifiable directly: $\text{AIN}(\mathbf{a}, \mathbf{b})$ with no third argument defaults to $c = (a + b)/2$, yielding $\alpha = \beta$ and reproducing classical interval behavior.

The degenerate case $a = b = c$ is handled separately: the density representation is bypassed and all operations reduce to standard real arithmetic, consistent with the Dirac distribution at c described in Section 2.4.

Beyond the operations listed in Table 1, the LOTUS mechanism extends naturally to any transformation with a known antiderivative. Version 1.2 [6] exploits this by implementing trigonometric functions (sine, cosine, tangent), the exponential function, and logarithms (natural, base-2, base-10), each with analytic expected-value propagation and explicit handling of domain restrictions. The library also provides logical predicates (degeneracy, positivity, symmetry), relational operators (overlap, subset, disjointness), construction of AIN objects from empirical samples, and normalization of AIN collections. All operations maintain the same principle: bounds from the range of the transformation, expected value from LOTUS. A complete description of the API is given in the documentation website⁴.

5 Properties of the AIN Distribution

Because the AIN induces a fully specified probability distribution, all standard distributional characteristics can be expressed in closed form. It should be emphasized that the induced distribution is a canonical information-consistent representation of the available knowledge (a, b, c) , not a claim about the physical data-generating process. The properties derived below describe the logical consequences of this representation. Throughout, let $X = [a, b]_c$ be a non-degenerate AIN with density parameters α and β given by Eq. (9).

5.1 Distribution Function, Quantiles, and Median

Integrating the density (6) yields the cumulative distribution function:

$$F_X(x) = \begin{cases} 0, & x < a, \\ \alpha(x - a), & a \leq x < c, \\ \alpha(c - a) + \beta(x - c), & c \leq x < b, \\ 1, & x \geq b. \end{cases} \quad (14)$$

The CDF is continuous and piecewise linear with slopes α on $[a, c)$ and β on $[c, b]$. It reaches the value

$$F_X(c) = \alpha(c - a) = \frac{b - c}{b - a} \quad (15)$$

⁴ <https://asymintervals.readthedocs.io/en/latest/>

at the expected value. Note that $F_X(c)$ depends only on the relative position of c within $[a, b]$, not on the absolute values of a , b , or c .

The quantile function (inverse CDF) is obtained by inverting each linear piece:

$$Q_X(p) = \begin{cases} a + \frac{p}{\alpha}, & 0 \leq p \leq F_X(c), \\ c + \frac{p - F_X(c)}{\beta}, & F_X(c) < p \leq 1. \end{cases} \quad (16)$$

The median is the quantile $Q_X(1/2)$. The condition $F_X(c) \geq 1/2$ is equivalent to $c \leq (a + b)/2$, so the median lies in the left subinterval $[a, c]$ when the expected value is in the lower half of the interval, and in the right subinterval $[c, b]$ otherwise. In the symmetric case $c = (a + b)/2$, the median coincides with the expected value and the midpoint.

5.2 Variance and Entropy

The second moment of X under the induced density is

$$\mathbb{E}(X^2) = \alpha \frac{c^3 - a^3}{3} + \beta \frac{b^3 - c^3}{3}. \quad (17)$$

Substituting the expressions for α and β from (9) and simplifying yields a compact formula for the variance.

Proposition 1. *The variance of an AIN $X = [a, b]_c$ is*

$$\text{Var}(X) = \frac{(c - a)(b - c)}{3}. \quad (18)$$

Proof. Using the factorizations $c^3 - a^3 = (c - a)(c^2 + ac + a^2)$ and $b^3 - c^3 = (b - c)(b^2 + bc + c^2)$, the second moment becomes

$$\mathbb{E}(X^2) = \frac{(b - c)(c^2 + ac + a^2)}{3(b - a)} + \frac{(c - a)(b^2 + bc + c^2)}{3(b - a)}.$$

Expanding and collecting terms gives $\mathbb{E}(X^2) = (-ab + ac + bc + 2c^2)/3$. Since $\text{Var}(X) = \mathbb{E}(X^2) - c^2$, we obtain

$$\text{Var}(X) = \frac{-ab + ac + bc + 2c^2}{3} - c^2 = \frac{-ab + ac + bc - c^2}{3} = \frac{(c - a)(b - c)}{3},$$

where the last equality follows from $(c - a)(b - c) = -ab + ac + bc - c^2$. $\square$

The formula $\text{Var}(X) = (c - a)(b - c)/3$ depends only on the distances from c to the two endpoints and is maximized when $c = (a + b)/2$, where it reduces to $(b - a)^2/12$, the variance of the uniform distribution $\mathcal{U}(a, b)$.

The differential entropy of the induced distribution is

$$H(X) = - \int_a^b f_X(x) \ln f_X(x) dx = - \frac{b - c}{b - a} \ln \alpha - \frac{c - a}{b - a} \ln \beta. \quad (19)$$

Table 2. Properties of the AIN distribution for $X = [a, b]_c$ with $a < c < b$.

Property	Expression
Expected value	c
Variance	$(c - a)(b - c)/3$
Standard deviation	$\sqrt{(c - a)(b - c)/3}$
CDF at c	$(b - c)/(b - a)$
Entropy	$-\frac{b - c}{b - a} \ln \alpha - \frac{c - a}{b - a} \ln \beta$
Asymmetry coefficient	$(a + b - 2c)/(b - a)$

In the symmetric case $c = (a + b)/2$, this reduces to $H(X) = \ln(b - a)$, the entropy of $\mathcal{U}(a, b)$. For asymmetric AINs, the entropy is strictly less than $\ln(b - a)$, reflecting the additional information encoded in the expected value c .

Remark 2. Among all absolutely continuous distributions on $[a, b]$ with expected value c , the distribution maximizing the differential entropy is the truncated exponential density $f(x) \propto e^{\lambda x}$, where λ is determined by the expectation constraint (reducing to the uniform distribution when $c = (a + b)/2$). The AIN density does not maximize entropy in this general sense. It does, however, maximize entropy within the class of piecewise-constant densities with breakpoint at c , since the uniform distribution on each subinterval is the maximum-entropy distribution on that subinterval given no further constraints. The AIN construction is thus motivated not by global entropy maximization but by the principle of indifference applied separately to each subinterval induced by c .

Table 2 collects the distributional properties derived in this section. All properties are determined by the triple (a, b, c) alone and reduce to the corresponding properties of $\mathcal{U}(a, b)$ when $c = (a + b)/2$.

6 Further Directions

In the approach considered here, the joint distribution is taken to be the product of the marginal densities (Section 3). However, when information about the dependence between the variables is available, such as a prescribed correlation or a copula, this assumption becomes inadequate. Constructing an AIN arithmetic under specified dependence structures would require parameterizing a joint density while preserving the AIN marginals and constitutes a major open problem.

In the present formulation, the parameters (a, b, c) are assumed given. When observations are available, one may ask how these parameters should be estimated from a sample. The parameter c is naturally estimated by the sample mean, while the endpoints a and b are more delicate and may be identified either with the sample extrema or with more robust, for example quantile-based, bounds. A coherent estimation methodology and its statistical analysis are still lacking.

The AIN arithmetic also admits extensions to additional nonlinear transformations, including exponentiation with scalar and interval-valued exponents. The quantile representation (16) furthermore induces natural distance measures between AINs, as the closed-form quantile function makes Wasserstein-type distances available in explicit form. A detailed treatment of these constructions is currently in preparation and will be presented elsewhere.

Another natural direction concerns the use of AINs in comparison and ranking problems. In many decision contexts, uncertain quantities must be compared. An AIN-based comparison of two quantities X and Y does not produce a binary order, but rather an uncertain preference that can itself be represented as an AIN. This suggests the construction of preference graphs in which the nodes represent quantities described by AINs, while the edges encode the preference relations between them, capturing both the direction and the uncertainty of each pairwise comparison. Such structures provide a more informative representation of ranking uncertainty than scalar aggregation and may form a basis for uncertainty-aware decision models. A systematic study of this approach is currently in preparation and will be presented elsewhere.

Finally, the current framework is univariate. For applications involving multiple uncertain quantities with known joint structure, a multivariate AIN formulation would be needed, raising questions about the representation of joint information and the tractability of operations in higher dimensions.

7 Conclusions

AINs have a natural probabilistic interpretation. Starting from the classical interval and the principle of indifference, the induced piecewise-uniform density is uniquely determined by the normalization and expectation constraints and contains no free parameters beyond the triple (a, b, c) .

The piecewise-constant class is itself motivated by the principle of indifference applied to each subinterval, extending the same reasoning that associates the uniform distribution with a classical interval. Within this class, the density is not a modeling decision but a logical consequence of the constraints. The expected value c is treated as a primitive parameter encoding partial knowledge, on a par with the bounds a and b , rather than as a derived moment of an assumed distribution. Consequently, the AIN should not be interpreted as one uncertainty model among many, but as the minimal probabilistic extension of classical interval information when an expected value is available.

Within this framework, all standard operations, from basic arithmetic to distributional characteristics such as variance, entropy, and quantile functions, admit explicit analytical expressions. This eliminates the sampling noise inherent in Monte Carlo propagation and guarantees deterministic, reproducible results, as confirmed by the reference implementation in the `asymintervals` library. Deterministic propagation does not eliminate the underlying epistemic uncertainty; it provides a specific, consistent representation of that uncertainty within the adopted framework.

The AIN representation is most beneficial when a practitioner possesses knowledge of an expected or typical value in addition to bounds, yet the exact distribution is unknown or too costly to determine. Such situations arise frequently in engineering tolerances, cost estimation, and expert elicitation. In these settings, the AIN provides the minimal consistent representation of uncertainty between classical intervals and fully specified probability distributions. Conversely, when the underlying distribution is known with confidence, or when the expected value coincides with the midpoint, the classical interval framework remains adequate, and the AIN reduces to it by Theorem 2.

Several directions remain open, including AIN arithmetic under specified dependence structures, coherent estimation of the triple (a, b, c) from data, and formal connections to imprecise probabilities and Dempster–Shafer theory. These directions indicate that AINs may serve as a bridge between interval-based and probabilistic representations of uncertainty, suggesting a broader role in uncertainty-aware modeling and decision analysis.

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